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Robotic Tissue Manipulation in Endoscopic Submucosal Dissection via Visual Feedback

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***Abstract*— Colorectal cancer is the third most commonly diagnosed cancer and the second leading cause of cancer-related deaths in the United States. Despite advancements in screening and treatment, there remains a critical need for more effective and minimally invasive methods to manage complex polyps and early-stage colorectal cancers. This study introduces a novel approach to magnetic tissue manipulation for Endoscopic Submucosal Dissection (ESD), leveraging visual feedback to enhance precision and control. We develop and evaluate the proposed system within a ROS Gazebo simulation environment, integrating a small magnetic endoscopic clip affixed to tissue, which is manipulated by an external large magnet mounted on a robotic arm.**

A key challenge in ESD is achieving adequate tissue exposure for precise cutting, particularly in the confined space of the colon where the endoscope is manually controlled. To address this, our system enables controlled manipulation of the magnetic clip to optimize tissue retraction. The robotic arm, guided by real-time visual feedback, dynamically adjusts the internal clip's orientation. Multiple virtual cameras were used to validate the proposed method. The simulation results demonstrated that the robot arm successfully manipulated the internal magnetic clip to the desired tilt angle within an average of 8.4 seconds (range 5.3 to 15.2 s). Our findings suggest that robotic-assisted magnetic tissue manipulation has the potential to improve ESD success rates while reducing procedure time, paving the way for further advancements in minimally invasive endoscopic surgery.

I. INTRODUCTION

The application of magnetic interaction for surgical procedures has demonstrated success [1] in areas such as diagnostics [2], targeted drug delivery [3], and robotic surgery, particularly when applied to manipulation of tissue in constrained spaces in the body with limited access for maneuverability [4]. A major reason for this is the constrained or obstructed operational spaces that prevent linked devices from passing through or approaching without interference from other essential tools. Application of magnetic force provides the opportunity to manipulate instruments that are not directly tethered to a control outside of the body, thereby broadening the potential range of maneuverability within a small, confined space.

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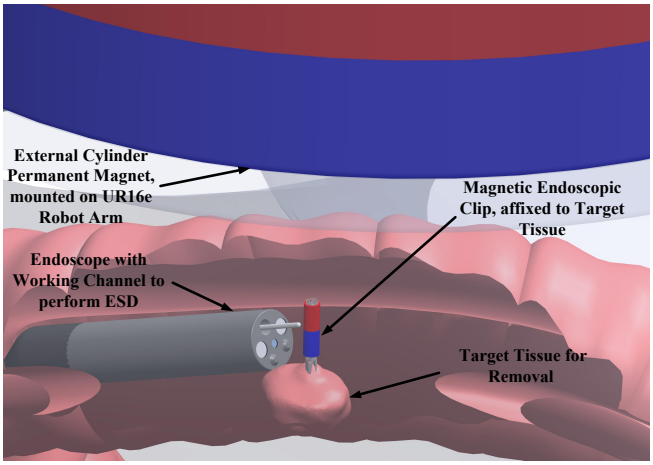


Fig. 1. The magnetic manipulation technique studied involves an external magnet mounted on a robotic arm which directs an internal magnet incorporated into an endoscopic clip affixed to tissue undergoing endoscopic resection.

Magnetic manipulation typically works through the combination of an external magnetic source and an internal magnetic object. The external magnetic source actively controls the motion by changing the magnetic field, while the internal magnetic object adjusts its pose (position and/or orientation) to align its magnetic field with the external one. From early-stage exploration to recently reported clinical trials, the manipulation of external magnetic sources has largely been achieved through handheld methods [4]–[6]. This requires specific medical staff to assist, communicate with the surgeon, and manipulate the magnetic source.

Traditional Endoscopic Submucosal Dissection (ESD) procedures consist of four phases including marking, injection, mucosal incision, and submucosal dissection. In our and reported data of unassisted ESD, dissection was the most time-consuming aspect of the procedure, requiring an average of 53.85% of the total time or $1,394.7$ seconds out of $2,590$ seconds [7], as successful and safe submucosal dissection requires accurate cutting of tissue on the order of mm. The entire submucosa underneath the targeted tissue must be dissected free. We introduced magnetic retraction of the targeted tissue to facilitate ESD, effectively introducing an additional "hand" that is not physically tethered to a system outside of the body. The absence of a physically attached tether in addition to the fine manipulation achieved by the external magnet provides fine-tuned, readily adjustable control of retraction of tissue to expose a safe plane of submucosal dissection and incision of the targeted lesion for safe and efficient excision. We describe experiments using

both ex-vivo and live animal models where the ROS Gazebo model is successfully applied and decreases ESD time.

In recent years, various tissue manipulation techniques have been developed. Some are manually controlled with magnetic assistance [4], [6], [8], [9], some are robotically controlled without magnetic assistance [10]–[12], and others are robotically controlled with magnetic assistance [13], each targeting different applications.

In [8], surgeons compared the efficacy of ESD using a novel magnetic device with the traditional technique. Their trials demonstrated the benefits of the magnetic method in terms of directional tissue manipulation and exposure of the submucosal space. They recorded an average time reduction of around 44% (605 ± 303.7 vs $1,082 \pm 515.9$ seconds), significantly reducing procedure time compared with conventional ESD techniques. In the reported procedure, no robotic devices were involved, and the external magnet was manually adjusted.

In [9], surgeons reported 50 clinical trials using the magnetic surgical system (Levita Magnetics' Surgical System) during reduced-port laparoscopic cholecystectomy. They successfully performed procedures reported no device-related serious adverse events and achieved a high level of acceptance (90% "excellence") by the surgeons, although the external cylindrical magnet was handheld to manipulate the detachable magnetic grasper secured to the organ.

In [10], two machine learning-based model predictive control algorithms with vision strategies achieved automated tissue manipulation. They conducted experiments by adapting the learning from demonstration (LfD) algorithm on a Raven IV platform.

In [4], surgeons reported clinical comparisons between using magnetic liver retraction (same commercial system as used in [9]) and bedrail-mounted liver retractors in terms of pain and resource usage across hundreds of bariatric surgery cases. They confirmed the positive impact of magnetic-assisted devices, including reduced postoperative pain, and shorter hospital stays. The external magnet's motion was also manually controlled.

In [6], surgeons and researchers presented an animal study on magnetic-assisted thoracoscopic lobectomy. Their reported data- including postoperative blood loss, surgical complications, and the effect of intraoperative device use- verified the safety and feasibility of magnetic-assisted tissue manipulation in thoracoscopic surgery. The external magnet was controlled manually by medical staff.

In [11], the authors presented a sim-to-real framework for automating tasks in a simulation environment, planning ahead to find grasping points that minimize local tissue deformation by combining deep reinforcement learning and FEM simulation. Experimental trials validated the proposed sim-to-real method's effectiveness, showing that learned policies could accurately place internal tissues with minimal tissue deformation during planned grasping operations. However, the reported average planning time of around 377 seconds remains a challenge for tissue/organ manipulations requiring real-time, fast-convergence control.

TABLE I
COMPARISON OF CURRENT TISSUE/ORGAN MANIPULATION TECHNIQUES.

Ref.	Year	Mount Method	Sensing	Manipulation Method
[8]	2014	Hand-held	Surgeon	Manual Motion
[9]	2018	Hand/on Device	Surgeon	Manual Motion
[4]	2021	Hand/on Device	Surgeon	Manual Motion
[6]	2022	Hand-held	Surgeon	Manual Motion
[10]	2019	Raven IV	Camera	MPC+LfD
[11]	2023	DVRK PSMs	Camera	DRL
[12]	2024	DVRK PSMs	Camera	Shared control
[13]	2024	Robot Arm	Unknown	Robotic

In [12], the authors reported a robotic framework for manipulation and cutting unknown deformable tissue. Their method utilized online data-driven learning, adaptive control, and a reduced-order trajectory planning module. They implemented this combined approach on the dVRK©platform and performed 25 experiments on silicon phantoms, demonstrating significant progress in dynamic tissue manipulation and cutting.

In [13], the magnetic system reported in [4], [9], which featured manual control of the external magnet, evolved into robotic control with the introduction of dedicated robot arms. Surgeons assessed the safety and efficacy of urologic procedures using the new magnetic-assisted robotic surgery (MARS©) platform. They confirmed that the robotic-controlled external magnet significantly aided tissue retraction without requiring additional incisions, reduced the need for extra medical staff, enhanced surgeon control and visualization, and accelerated the learning curve.

Here, we compare the above tissue/organ manipulation methods, both manual and robotic, in Table I, to better understand the features of these techniques. It can be seen that the magnetic-assisted tissue/organ manipulation and levitation techniques have evolved from initial manual control to intelligent robotic control. Meanwhile, tissue manipulation techniques developed without magnetic assistance, using more complex algorithms, take more time to reach targets for relatively short trajectories, and remain in the theoretical phase without in-vivo clinical trials. However, these developed robotic manipulation methods still provide essential approaches for surgery scenarios where magnetic assistance is not suitable or unable to handle complex manipulation tasks. In reality, technical challenges remain for both magnetic-assisted and non-magnetic-assisted manipulations. For instance, surgical instruments introduced into the body (e.g., the endoscope) can be pulled by the external magnet if the distance is too close, even though the benefits of magnetic-assisted tissue/organ levitation and manipulation are apparent and widely acknowledged. Additionally, a robotic approach offers the opportunity for safety incorporation that would

avoid the human error of incorrect magnet placement. Lastly, it is worth noting that many other tissue/organ manipulation methods report valuable results but are not covered by this section's review.

The main contributions of this work lie in several aspects. First, the successful implementation of the ROS-Gazebo simulation environment, along with the demonstration of vision feedback-based magnetic robotic manipulation, provides a feasible and validated approach for similar scenarios. Second, through multiple simulation trials, we demonstrated that the tissue clamp can be adjusted in just a few seconds, paving the way for more clinically relevant ex-vivo and in-vivo trials, including reducing the dependence on the technical staff for endoscopists. Third, we provide a possible approach to enable conventional endoscopes to be equipped with automation and intelligence by integrating the endoscope's visual feedback into the ROS platform.

The structure of this paper is arranged as follows. Section II introduces the ROS Gazebo simulation environment, including the customized magnets and YOLO-based magnetic clip recognition. Section III presents the methods employed in this paper, including magnetic force and torque calculations, YOLO detected magnetic clip's tilt angle calculation, and the steepest gradient method-based navigation algorithm. Section IV demonstrates the results obtained from simulation trials with multiple camera fields of view. Limitations and challenges for real-world implementation are also discussed in this section. Section V concludes the paper and outlines plans for the next steps in ex-vivo and in-vivo studies.

II. ROS GAZEBO SIMULATION ENVIRONMENT

In this section, we describe the establishment of the ROS Gazebo-based simulation environment, including the configuration of external and internal magnets, as well as the detection of the magnetic clip and calculation of the tilt angle.

A. Basic ROS Gazebo Configurations

The simulation environment is based on the publicly available ROS package and driver for the Universal Robot Arm UR16e. The surgery table, robot arm mount, tissue platform, and other components are configured using URDF formats. The configuration of the magnetic clip is based on recorded video from the endoscopic camera's field of view (FoV) during the team's previous in-vivo ESD experiments on animals. In this study, we only focused on the motion of the clip in relation to the external magnet.

B. Magnetic Configuration

The permanent magnets used in this simulation study were based on a real, manufactured, and experimentally tested pair. A cylindrical N52 NdFeB magnet with a diameter of 210 mm and a thickness of 50 mm was used as the external magnet, while N55 NdFeB magnet with an inner diameter of 2.1 mm, an outer diameter of 3.1 mm, and a thickness of 10 mm was used as the internal magnet.

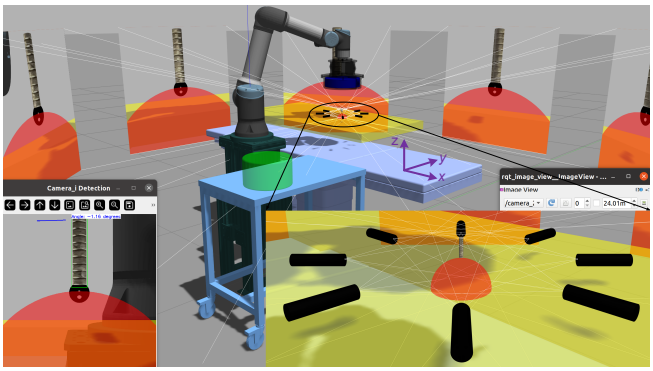


Fig. 2. The established ROS Gazebo Simulation environment.

C. Magnetic Clip Recognition

We utilize YOLO (You only look once) to detect the magnetic clip and calculate its angle in real time. For labeling purposes, we ran several simulations in Gazebo and collected over two thousand images during these simulations. The collected images were manually labeled and used for offline training to obtain a model for later online real-time inference, aimed at detecting the magnetic clip using the YOLO V5 model [14].

The established simulation environment is shown in Figure 2. In order to mimic real clinical scenarios, where the endoscopic camera changes its position and orientation randomly, we placed eight cameras surrounding the magnetic clip to observe and provide feedback images for real-time recognition and angle calculation during the subsequent simulation trials. It is worth noting that for each simulation trial, only the uniquely selected camera's view will be used as visual feedback for the proposed algorithm.

III. METHODS

A. Magnetic Interaction in ROS Gazebo Framework

The magnetic interaction between the external and internal magnets within our established ROS Gazebo simulation environment is based on the dipole approximation for magnetic field-induced force and torque.

The magnetic force between two dipoles $\mathbf{m}_1$ and $\mathbf{m}_2$ separated by a distance vector $\mathbf{r}$, is given by:

$$\mathbf{F} = \frac{3\mu_0}{4\pi r^5}\Big[(\mathbf{m}_1 \cdot \mathbf{r})\mathbf{m}_2 + (\mathbf{m}_2 \cdot \mathbf{r})\mathbf{m}_1 + (\mathbf{m}_1 \cdot \mathbf{m}_2)\mathbf{r} - \frac{5(\mathbf{m}_1 \cdot \mathbf{r})(\mathbf{m}_2 \cdot \mathbf{r})}{r^2}\mathbf{r}\Big] \tag{1}$$

Where:
- μ_0 is the magnetic permeability of free space.
- $r = |\mathbf{r}|$ is the distance between the dipoles.
- $\mathbf{m}_1$ and $\mathbf{m}_2$ are the magnetic dipole moments.

The magnetic torque experienced by the dipole $\mathbf{m}_1$ in a magnetic field $\mathbf{B}_2$ generated by $\mathbf{m}_2$ is:

$$\mathbf{T} = \mathbf{m}_1 \times \mathbf{B}_2 \tag{2}$$

The magnetic field $\mathbf{B}_2$ at a point due to the dipole $\mathbf{m}_2$ is:

$$\mathbf{B}_2 = \frac{\mu_0}{4\pi r^3}\left[3(\mathbf{m}_2 \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}_2\right] \tag{3}$$

Algorithm 1 Tilt Angle Calculation using PCA method

Input: Cropped image cv_image_crop
Convert the image to grayscale: $gray \leftarrow$ cv2.cvtColor(cv_image_crop, cv2.COLOR_BGR2GRAY)
Apply binary thresholding: $binary \leftarrow$ cv2.threshold($gray, 127, 255$, cv2.THRESH_BINARY)
Extract coordinates of non-zero pixels: $points \leftarrow$ np.column_stack(np.where(binary > 0))
if len($points$) > 0 **then**
  Convert $points$ to float32: $points \leftarrow$ np.array($points, dtype = np.float32$)
  Perform PCA: mean, eigenvectors $\leftarrow$ cv2.PCACompute($points$, mean = np.array([]))
  Compute the angle:

$$\text{angle} \leftarrow \arctan\left(\frac{\text{eigenvectors}[0,1]}{\text{eigenvectors}[0,0]}\right) \times \frac{180}{\pi}$$

  Output: The calculated angle in degrees
end if

Where:
- $\hat{\mathbf{r}} = \frac{\mathbf{r}}{r}$ is the unit vector in the direction of $\mathbf{r}$.

We implemented the above equations and the magnet parameters introduced in Section II-B into a customized ROS package to achieve the magnetic manipulation simulation. We also refer to this work [15] while establishing the simulation environment for magnetic interaction.

B. YOLO based Magnetic Clip's Angle Calculation

Based on the YOLO-detected magnetic clip, we apply the following steps for calculating the angle using Principal Component Analysis (PCA):

1. Grayscale Conversion: Convert the RGB image to grayscale.

2. Binary Thresholding: Apply binary thresholding to isolate the object of interest.

3. Extract Non-zero Points: Extract the coordinates of non-zero pixels (i.e., foreground pixels).

4. Principal Component Analysis (PCA): - First, represent the extracted points as a matrix $\mathbf{X} \in \mathbb{R}^{n\times 2}$ where each row represents a point (x_i, y_i). - Then, compute the mean and eigenvectors.

5. Angle Calculation: - The angle θ between the first principal eigenvector and the x-axis can subsequently be calculated.

Detailed algorithms for this angle calculation are presented in Algorithm 1.

C. Steepest Gradient-based Navigation Algorithm

Due to the nature of magnetic field-based interactions, the internal magnetic clip will always try to adjust its orientation until its magnetic field aligns with the external one. To simplify the manipulation of the internal magnetic clip's pose, the external magnet only needs to move along a plane that maintains a fixed vertical distance (a distance that prevents the patient from being compressed by the external magnet and ensures the endoscope is not interfered by the external magnetic field) from the internal magnet. In this study, we kept the orientation and z-position of the external magnet fixed, while only navigating it along the xy-plane.

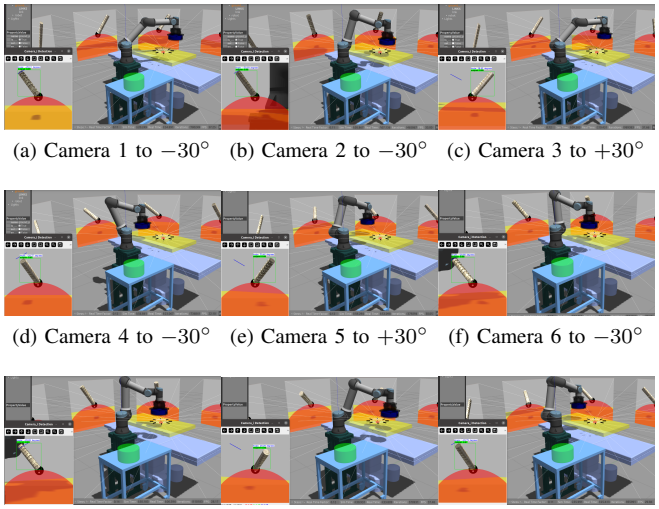
A steepest gradient-based navigation algorithm was designed and tested. Initially, we tested the standard steepest gradient method to manipulate the magnetic clip's angular

Algorithm 2 End Effector Trajectory Navigation Algorithm

Initialize:
Read $P_0 = (x_0, y_0, z_0)$ from position sensor.
Move end effector above P_0.
Calculate:
Define circle centered at (x_0, y_0) with radius r.
Calculate 12 coordinates P_1 to P_{12} on the circle at 30-degree intervals:

$$P_1 = (x_0 + r, y_0)$$
$$P_2 = \left(x_0 + r\cos\left(\frac{\pi}{6}\right), y_0 + r\sin\left(\frac{\pi}{6}\right)\right)$$
$$\vdots$$
$$P_{12} = \left(x_0 + r\cos\left(\frac{11\pi}{6}\right), y_0 - r\sin\left(\frac{11\pi}{6}\right)\right)$$

Trial:
for $i = 1$ **to** 12 **do**
  Move end effector to P_i.
  Pause for 0.5 seconds.
  Read and store tilt angle.
  if tilt angle reaches trigger value **then**
    Skip remaining trials.
    Use P_i's direction as the desired direction.
    Break.
  end if
  Move end effector back to P_0.
end for
Choose Trajectory:
if tilt angle reached trigger value during trials **then**
  Use the direction from the triggered position P_i.
else
  Compare tilt angles and identify two coordinates with the largest and second-largest error reduction.
  Calculate the direction between these two coordinates.
end if
Execute:
Move end effector along the calculated direction.
Monitor tilt angle in real-time.
Adjust navigation speed based on error reduction:
if error reduced by 85% **then**
  Reduce speed.
else if error reduced by 95% **then**
  Further reduce speed.
end if
if tilt angle error $<$ threshold (e.g., 1 or 2 degrees) **then**
  Stop program.
end if



(a) Camera 1 to $-30°$ (b) Camera 2 to $-30°$ (c) Camera 3 to $+30°$

(d) Camera 4 to $-30°$ (e) Camera 5 to $+30°$ (f) Camera 6 to $-30°$

(g) Camera 6 to $-30°$ (h) Camera 7 to $+30°$ (i) Camera 8 to $-30°$

Fig. 3. Screenshots from the ROS Gazebo simulation showing the manipulation of the internal magnetic clip's tilt angle.

position. However, we found that this algorithm was not suitable for our application, as it attempted to find the steepest direction, which either took a long time or led to instability when the learning rate was increased. Due to the nature of the application scenario, the magnetic clip does not need to be manipulated in the steepest direction (i.e., associated tilt in other directions is allowed). A variant of the steepest gradient method was proposed, employing a try-and-execute approach. The end effector navigates an equal distance in different directions along a circle centered at the magnetic clip's location, with equal angular intervals. The starting direction is randomly generated. We conducted multiple pre-trials under each camera's field of view to determine the angular value used to trigger the "direction identified" event. In the formal experiments, once a direction results in a significant change (absolute value, 25%) in the magnetic clip's angle towards the predefined value, the algorithm skips the remaining trials and triggers navigation of the end effector toward that direction or its reverse, providing real-time feedback on the angular position of the magnetic clip, until the error between the detected and desired angles is smaller than a predefined value. The detailed algorithm is demonstrated in Algorithm 2.

IV. RESULTS AND DISCUSSION

A. Results

Based on previously conducted ex-vivo and in-vivo ESD trials involving manual manipulation of the internal magnet orientation, we identified an angular range between 11 O'clock($-30°$) and 1 O'clock($+30°$) that covers most scenarios necessary to facilitate the dissection task. Consequently, we set the tilt limit to ($\pm 30°$) and obtained the simulated results, as shown in Figure 3. A lookup table summarizing the maximum and average time cost for each case and the target value is shown in Table II. The proposed algorithm can navigate the robot arm to manipulate the magnetic clip's tilt angle from an initial state (zero degrees) to a desired angle of up to $\pm 30°$, all within the field of view of any randomly placed endoscopic camera, in approximately 8.4 seconds, with a minimum of 5.3 seconds and a maximum of 15.2 seconds.

TABLE II

SIMULATION RESULTS OF MAGNETIC CLIP TILT MANIPULATION UNDER DIFFERENT CAMERAS AND TARGETS.

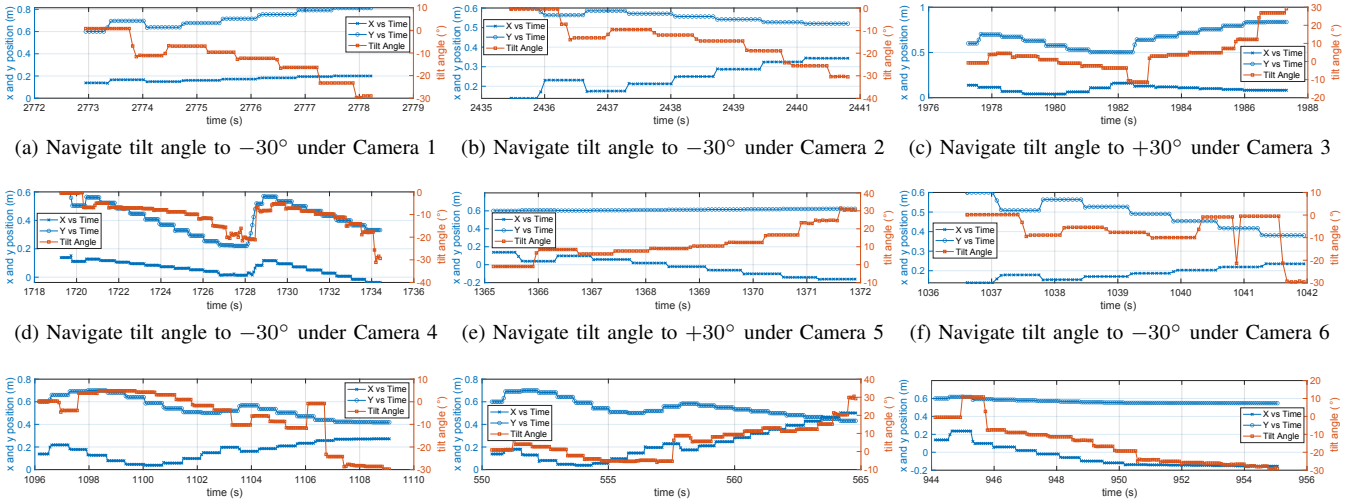
Camera ID	Target angle (°)	Max. (s)	Min. (s)	Average (s)
1	-30	9.4	5.4	6.76
	+30	8.73	5.4	7.19
2	-30	10.3	5.33	8.47
	+30	12.2	6.3	9.13
3	-30	11.7	7.5	9.32
	+30	10.3	6.1	8.34
4	-30	15.2	6.1	9.18
	+30	12.4	6.1	8.9
5	-30	6.8	5.4	5.94
	+30	8.2	6.8	7.08
6	-30	13.1	5.33	9.31
	+30	11.0	7.4	9.43
7	-30	11.7	6.1	9.15
	+30	14.5	5.4	8.76
8	-30	11.0	7.4	8.74
	+30	10.3	5.93	8.73
1∼8	±30	**15.2**	**5.3**	**8.4**

We also plotted the trajectories of the robot arm's end effector for selected tasks, including repeated trials, as presented in Figure 4 and Figure 5 (trajectories with the maximum and minimum time used for each case, for comparison if repeated). It is worth noting that for the same tasks, such as navigating the magnetic clip's tilt angle to $-30°$ under camera 6, the executed trajectories differ, with shorter trajectories representing shorter convergence times.

Compared with the ex-vivo and in-vivo trials previously conducted by our team, the simulated results show a reduction in the time needed to tilt the magnetic clip's angle to the desired angular position from minutes to seconds, demonstrating promising potential for real-world implementation. Meanwhile, it is estimated that if the tilt limit increases to $\pm 45°$, the time required will only increase slightly, keeping the total operation time within an acceptable range.

B. Limitations

The detection and calculation of the magnetic clip are based on simulations where no random noise exists in the camera's field of view. This is inconsistent with real clinical scenarios, where patient respiration movements, variations in distance depending on the location of the targeted tissue, differences in body habitus, and how these factors affect the distances between external and internal magnets, as well as the external maneuverability of the magnet around the body, must be considered. Additionally, other minor issues including low visibility can occur due to factors such as light reflection from tissue, minor stains or fog on the lens, the camera constantly changing and making it difficult to obtain a suitable field of view, or camera shaking/oscillation when hand-held by the surgeon. These issues contribute to noise



(a) Navigate tilt angle to $-30°$ under Camera 1 (b) Navigate tilt angle to $-30°$ under Camera 2 (c) Navigate tilt angle to $+30°$ under Camera 3

(d) Navigate tilt angle to $-30°$ under Camera 4 (e) Navigate tilt angle to $+30°$ under Camera 5 (f) Navigate tilt angle to $-30°$ under Camera 6

(g) Navigate tilt angle to $-30°$ under Camera 6 (h) Navigate tilt angle to $+30°$ under Camera 7 (i) Navigate tilt angle to $-30°$ under Camera 8

Fig. 4. Positions of the external magnet mounted on the UR16e robot arm's end effector and the angles of the internal magnetic clip, along with timestamps, were obtained from the ROS Gazebo simulation. Selected results are presented in this figure.

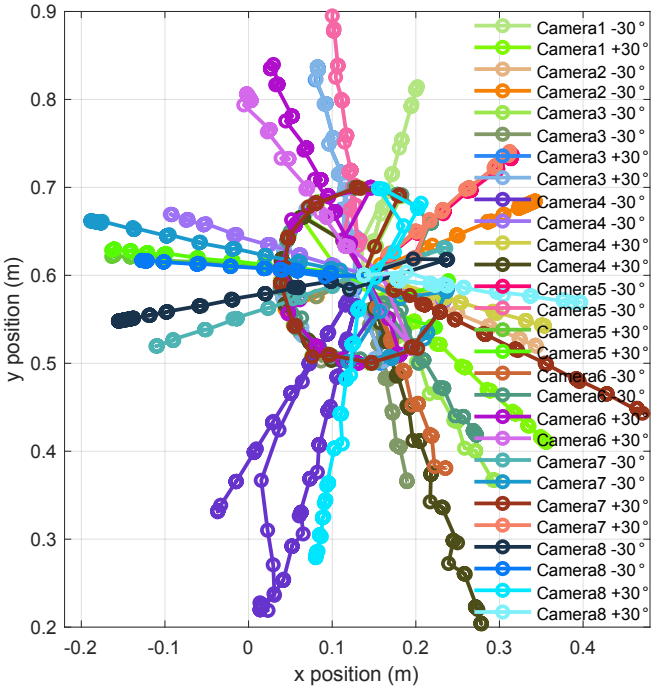


Fig. 5. The trajectories of the UR16e's end effector, executed to navigate the magnetic clip, achieving different target tilt angles under various cameras.

that affects the detection of the magnetic clip and tilt angle calculation.

In the simulation, the magnetic clip is clamped to a rigid tissue, which does not reflect real scenarios where the tissue moves with the clip to uncover the submucosal layer. The simultaneous motion of the soft tissue causes the clip's tilt pose to behave differently than in the current simulation, necessitating modifications of the manipulation algorithm.

The algorithm used to calculate the detected magnetic clip's angle sometimes derived incorrect values due to various reasons. The angle of the magnetic clip currently used for feedback to the control algorithm is limited to the pitch angle. To enable the algorithm to determine a more optimal direction with shorter travel distance and time, we plan to calculate the clip's second angular parameter, the yaw angle, which should improve the overall manipulation results.

V. CONCLUSION AND FUTURE WORK

In this paper, we present a tissue manipulation method to navigate the magnetic clip's tilt angle to a desired value in order to expose the submucosal layer during the ESD procedure. We implemented and validated the proposed method in a customized ROS Gazebo simulation framework, using a simplified environment (assuming no patient or tissue motion, no variations in background brightness, and no lens stains or fog). The results presented in Figure 3, Figure 4, Table II, and Figure 5 show that the internal magnetic tissue clamp can be adjusted to the desired orientation ($\pm 30°$) within a few seconds (average: 8.4 seconds) based on endoscopic visual feedback and an external robotic manipulator, demonstrating the effectiveness of the method.

The next steps involve implementing mockup-based, ex-vivo, and in-vivo trials using the techniques developed in this paper to identify real-world problems and improve the robustness of magnetic clip detection, tilt angle calculation, and tilt manipulation algorithms under complex experimental conditions. The experiments may introduce low visibility, and camera movement due to a hand-held endoscopic camera, lens stain, and fog. Other control approaches, such as model predictive control, neural network-based learning methods, as well as hybrid approaches may also be explored.

The developed tissue manipulation system will help relieve the steep learning curve, enable inexperienced surgeons to master ESD procedures more easily and decrease the need for technical staff in robotic surgery. This could improve the accessibility of ESD, allowing more patients to benefit from this procedure for early-stage colorectal cancer treatment.

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